

HAMA, Syrian Arab Republic

WMO: 400300

Lat: 35.117N

Lon: 36.750E

Elev: 303

StdP: 97.74

Time Zone: 2.00 (SYR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	1.1	-5.9	2.4	5.7	-4.0	2.8	6.1	8.4	12.2	6.1	12.6	0.1	110	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.4	39.5	21.2	37.9	20.8	36.5	20.4	23.2	34.3	22.4	33.6	21.8	33.0	1.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	15.0	28.6	18.8	14.1	27.7	18.1	13.5	27.0	70.4	34.0	67.6	34.1	65.3	33.0	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	5.9	4.6	-3.0	42.5	2.5	1.8	-4.8	43.7	-6.3	44.8	-7.7	45.7	-9.5	47.0		
(5)			-3.9	25.3	2.5	1.5	-5.7	26.3	-7.1	27.2	-8.5	28.0	-10.3	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	7.9	9.6	12.9	17.2	22.4	26.9	29.4	29.7	26.4	21.6	14.4	9.4		
	DBStd	8.24	2.64	2.91	2.96	3.44	3.06	2.39	1.85	1.92	2.17	3.20	3.36	2.85		
	HDD10.0	173	74	37	8	1	0	0	0	0	0	0	6	46		
	HDD18.3	1208	323	243	169	59	3	0	0	0	0	9	123	277		
	CDD10.0	3471	9	27	99	217	386	507	600	609	491	361	138	27		
	CDD18.3	1464	0	0	2	26	131	257	342	351	241	111	4	0		
	CDH23.3	17401	0	1	25	317	1462	3124	4458	4551	2528	902	33	0		
	CDH26.7	9260	0	0	5	97	654	1715	2591	2644	1235	316	2	0		
(14) Wind	WSAvg	1.1	0.6	0.9	1.1	1.3	1.4	1.8	1.9	1.4	1.1	0.6	0.4	0.3		
(15) Precipitation	PrecAvg	412	87	70	50	25	16	3	0	2	12	25	48	75		
	PrecMax	660	200	213	122	53	90	33	2	32	103	88	112	144		
	PrecMin	194	13	9	5	0	0	0	0	0	0	0	0	11		
	PrecStd	110	42	43	27	16	21	7	1	6	22	21	31	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	21.1	26.7	32.6	37.2	40.7	42.2	41.8	38.0	34.2	26.4	18.7		
		MCWB	11.7	13.4	14.7	18.4	18.9	20.5	21.4	21.9	20.8	18.6	15.7	12.5		
	2%	DB	15.5	18.3	23.0	29.2	34.9	37.9	39.5	39.3	35.7	31.9	24.0	16.8		
		MCWB	11.0	12.0	13.4	16.6	18.4	20.1	21.0	21.8	19.9	18.5	14.9	11.7		
	5%	DB	13.9	16.6	20.9	26.8	32.7	36.1	37.7	37.8	34.4	30.1	22.2	15.2		
		MCWB	10.2	11.2	12.9	15.7	17.6	19.3	20.8	21.2	19.6	18.0	14.2	11.2		
	10%	DB	12.4	15.0	19.1	24.6	30.6	34.4	36.1	36.2	32.9	28.3	20.5	13.8		
		MCWB	9.5	10.5	12.3	14.9	17.1	19.0	20.5	21.1	19.5	17.4	13.6	10.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	14.1	16.2	19.9	20.5	22.3	23.7	24.5	22.7	20.7	17.0	14.1		
		MCDB	16.2	20.0	23.4	29.6	31.7	35.3	36.0	35.6	32.7	29.5	23.7	16.7		
	2%	WB	11.6	12.6	14.6	17.7	19.3	21.2	22.7	23.6	21.7	19.6	16.0	12.7		
		MCDB	14.7	17.2	20.5	26.3	30.5	34.1	34.7	34.4	31.7	28.3	21.9	15.3		
	5%	WB	10.7	11.6	13.5	16.5	18.6	20.4	22.1	22.8	21.1	18.9	15.2	12.0		
		MCDB	13.3	15.7	19.6	24.9	29.8	32.9	34.2	33.6	30.8	27.5	20.5	14.4		
	10%	WB	9.8	10.7	12.6	15.5	17.8	19.8	21.5	22.2	20.5	18.3	14.3	11.2		
		MCDB	12.0	14.4	18.3	23.3	28.9	31.8	33.2	33.0	29.8	26.1	19.1	13.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.9	10.5	12.2	13.9	13.8	13.2	13.4	12.9	12.1	11.0	8.1		
		MCDBR	10.4	11.6	14.0	16.1	17.4	16.9	16.2	15.7	15.2	14.5	13.7	10.6		
		MCWBR	6.5	6.5	6.7	7.0	6.1	5.6	4.9	4.3	4.4	5.6	6.6	6.5		
	5% WB	MCDBR	8.7	10.5	12.5	14.6	14.9	14.5	13.4	13.8	12.9	12.4	11.0	8.1		
		MCWBR	5.7	6.2	6.5	6.9	5.7	5.2	4.5	4.2	3.8	5.0	5.6	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.368	0.395	0.430	0.484	0.473	0.420	0.425	0.437	0.420	0.438	0.385	0.371		
	taud		2.276	2.182	2.079	1.943	2.016	2.200	2.202	2.157	2.205	2.125	2.243	2.266		
	Ebn at Noon		815	834	835	808	821	864	857	839	836	776	786	783		
	Edh at Noon		104	128	154	185	175	145	144	148	134	134	107	100		
(44) All-Sky Solar Radiation	RadAvg		2.35	3.23	4.68	5.88	7.08	8.04	7.87	7.08	5.89	4.29	3.04	2.20		
	RadStd		0.21	0.37	0.29	0.38	0.39	0.16	0.14	0.16	0.21	0.22	0.24	0.25		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.30	N/A	-1.24	N/A	N/A	N/A	N/A	+173	+110

Nomenclature: See separate page